

Tranexamic acid: Drug information - UpToDate

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For additional information see "Tranexamic acid: Patient drug information" and "Tranexamic acid: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

Tranexamic Acid Safety Alert October 2025

The FDA is requiring labeling changes to strengthen the warnings about the risk of inadvertent intrathecal (spinal) administration of tranexamic acid injection. Tranexamic acid products must only be administered intravenously. The FDA took action after identifying and evaluating medication error cases where tranexamic acid was erroneously administered neuraxially (ie, intrathecal or epidural) instead of a local anesthetic, resulting in serious outcomes including prolonged hospitalization and death.

The FDA is requiring the following changes to the prescribing information for tranexamic acid injection: add a Boxed Warning about the risk of medication errors involving inadvertent neuraxial administration of tranexamic acid injection; add a statement to indicate that tranexamic acid injection is contraindicated as a neuraxial injection; clarify in Administration and Dosage that tranexamic acid injection must only be administered intravenously and provide instructions for preparing and administering the diluted solution.

Further information may be found at <https://www.fda.gov/drugs/drug-safety-and-availability/fda-provides-update-health-care-professionals-about-risk-inadvertent-intrathecal-spinal>.

ALERT: US Boxed Warning

Risk of medication errors due to incorrect route of administration (injection):

Tranexamic acid injection is for intravenous use only. Serious, including fatal, adverse reactions, including seizures and cardiac arrhythmias, have occurred when injectable tranexamic acid was inadvertently administered via the neuraxial route.

Brand Names: US

- Cyklokapron;
- Lysteda [DSC]

Brand Names: Canada

- Cyklokapron;
- Erfa-Tranexamic [DSC];

- GD-Tranexamic Acid;
- M-Tranexamic Acid;
- MAR-Tranexamic Acid

Pharmacologic Category

- Antifibrinolytic Agent;
- Antihemophilic Agent;
- Hemostatic Agent;
- Lysine Analog

Dosing: Adult

The adult dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editor: Edith A Nutescu, PharmD, MS, FCCP.

Dosage guidance:

Safety: Higher total IV doses (eg, ≥ 50 mg/kg), such as those given perioperatively, may be associated with an increased risk of seizures; lower doses (eg, 1 or 2 g given in the first 8 hours of trauma) do not appear to increase the risk of seizure or venous thromboembolism (Ref).

Abnormal uterine bleeding, acute

Abnormal uterine bleeding, acute (alternative agent) (off-label use):

Note: Hemodynamically unstable patients should generally be managed with surgical interventions (Ref). Alternative for patients without a high risk of thrombosis who fail other options or decline or should not use hormonal therapy (Ref).

Dosing: Multiple oral regimens have been described (Ref).

IV: 10 mg/kg (maximum 600 mg/dose) every 8 hours for up to 5 days (Ref).

Oral:

650 mg tablets: 1.3 **g** 3 times daily for up to 5 days (Ref).

500 mg tablets [Canadian product]: 1 to 1.5 **g** 3 times daily for up to 5 days (Ref).

Abnormal uterine bleeding, nonacute

Abnormal uterine bleeding, nonacute (alternative agent):

Note: Alternative as monotherapy for patients who cannot or choose not to use hormonal therapy. Initiate at the onset of heavy menstrual bleeding.

Oral:

Lysteda: 1.3 **g** 3 times daily for up to 5 days during monthly menstruation.

Cyklokapron [Canadian product]: 1 to 1.5 **g** 3 to 4 times daily for up to 5 days during menstruation (Ref).

Dental procedures in patients on oral anticoagulant therapy

Dental procedures in patients on oral anticoagulant therapy (off-label use):

Oral rinse: 5% solution (extemporaneously prepared): Administer 5 to 10 minutes prior to the procedure; hold 5 to 10 mL in mouth and rinse for 2 minutes; drain gently, being careful not to forcibly spit and dislodge clots; do not eat or drink for 1 hour after using oral rinse. Repeat 2 to 4 times daily for 1 to 2 days after the procedure (Ref).

Hemoptysis, treatment

Hemoptysis (nonmassive), treatment (off-label use):

Inhalation for nebulization: 500 mg (using injectable solution) 3 times daily for up to 5 days (Ref).

IV: 500 to 1,000 mg 3 times daily for 3 days (Ref).

Oral: 500 mg 3 times daily for 7 days or until hemoptysis is resolved (Ref); some experts recommend doses up to 1 g 3 times daily while awaiting definitive treatment (Ref).

Hereditary angioedema, long-term prophylaxis

Hereditary angioedema, long-term prophylaxis (off-label use):

Note: May be used for prophylaxis of hereditary angioedema (HAE) due to C1 inhibitor deficiency or HAE with normal C1 inhibitor; however, for patients with C1 inhibitor deficiency, other agents are preferred first line for prophylaxis (Ref).

Oral: Initial: 500 mg or 650 mg 2 or 3 times daily; titrate gradually based on response and tolerability up to a usual daily dose of 3 g/day in 2 or 3 divided doses; usual dosage range: 25 to 50 mg/kg/day (Ref).

Hereditary hemorrhagic telangiectasia, epistaxis or other bleeding sites

Hereditary hemorrhagic telangiectasia, epistaxis or other bleeding sites (alternative agent) (off-label use):

Note: May be used in carefully selected patients in whom local therapy and other management options are insufficient.

Oral: Initial: 500 mg twice daily or 3 g per day in 2 to 3 divided doses for 4 to 10 days; adjust dose as needed based on response and tolerability to a usual daily dose of 2 to 4.5 g in 2 to 4 divided doses (Ref).

Intracranial hemorrhage associated with thrombolytic treatment

Intracranial hemorrhage associated with thrombolytic treatment (alternative agent) (off-label use):

Note: Consider for use in addition to cryoprecipitate or when cryoprecipitate is contraindicated in patients who have a symptomatic intracranial hemorrhage after receiving thrombolytic within the past 24 hours (Ref).

IV: 1 g (or 10 to 15 mg/kg) once; administer at a rate not to exceed 100 mg/minute (generally over 10 to 20 minutes) (Ref).

Perioperative prevention of blood loss and transfusion

Perioperative prevention of blood loss and transfusion (eg, cardiac surgery, other surgeries with significant blood loss):

Note: There is wide variety in doses and routes of administration (IV, oral, and/or topical). Dosing and timing of administration are procedure and institution specific. Recommendations provided below are examples of IV regimens for use in selected surgeries; refer to institutional protocols.

Usual dose and range: IV: 1 g (or 10 to 30 mg/kg) prior to procedure; administer at a rate not to exceed 100 mg/minute (generally over 10 to 30 minutes). Depending upon type of procedure, a continuous infusion may be given intraoperatively after the initial bolus dose, or the bolus dose may be repeated at the end of procedure and/or during the postoperative period (Ref).

Cardiac surgery (off-label use):

Note: Optimal regimen is uncertain; refer to institutional protocol.

IV: Loading dose: 10 to 30 mg/kg administered at a rate not to exceed 100 mg/minute (generally over 10 to 20 minutes), followed by 1 to 16 mg/kg/hour (Ref). Alternatively, some centers administer a single loading dose of 50 mg/kg (Ref).

Orthopedic surgery (hip or knee arthroplasty) (off-label use):

Note: Optimal regimen is uncertain; refer to institutional protocol. Use in patients without a baseline high risk of thromboembolism. For patients with risk factors for thromboembolism, consider risk of thromboembolism vs benefit of reduced blood loss (Ref).

IV: 1 g (or 10 to 15 mg/kg) administered before skin incision at a rate not to exceed 100 mg/minute (generally over 10 to 20 minutes); repeat dose at skin closure or up to 12 hours later; some experts recommend a third dose during the postoperative period if needed (Ref). **Note:** Some experts use *intra-articular* tranexamic acid (ie, 1 g per 50 mL of NS applied topically into the wound at the end of the procedure) (Ref).

Spinal surgery (eg, spinal fusion) (off-label use):

Note: Optimal regimen is uncertain; refer to institutional protocol.

IV: 10 to 15 mg/kg administered prior to incision at a rate not to exceed 100 **mg/minute** (generally over 10 to 20 minutes), followed by 1 to 2 **mg/kg/hour** as a continuous infusion for the remainder of the surgery; discontinue at the end of the procedure (Ref).

Postpartum hemorrhage, prevention

Postpartum hemorrhage, prevention (adjunctive agent) (off-label use):

Note: For use in patients in high bleeding risk situations in conjunction with standard prophylactic uterotonics (eg, oxytocin) (Ref); some experts use in all patients regardless of bleeding risk (Ref).

IV: 1 g (or 10 to 15 mg/kg) over 10 to 20 minutes prior to skin incision (cesarean deliveries) or after cord clamping (cesarean or vaginal deliveries) (Ref). **Note:** Due to lack of robust data regarding fetal exposure, some experts administer after cord clamping regardless of birth route (Ref).

Postpartum hemorrhage, treatment

Postpartum hemorrhage, treatment (adjunctive agent) (off-label use):

Note: For continued bleeding despite oxytocin; used in conjunction with other therapies/procedures.

IV: 1 g over 10 to 20 minutes given within 3 hours of vaginal birth or cesarean delivery. If bleeding continues after 30 minutes, may repeat the dose in conjunction with thorough re-evaluation for cause of continued or recurrent bleeding (Ref).

Subarachnoid hemorrhage, prevention of early aneurysmal rebleeding

Subarachnoid hemorrhage, prevention of early aneurysmal rebleeding (off-label use):

Note: Routine use prior to aneurysm repair is not recommended (Ref). Optimal regimen has not been established; refer to institutional protocol.

IV: 1 g over 10 minutes immediately after diagnosis, followed by 1 g every 6 hours for no more than 72 hours (Ref). Alternatively, 1 g over 10 minutes immediately after diagnosis, followed by 1 g over 8 hours as a continuous infusion, continued until the start of endovascular or surgical treatment or until a maximum of 24 hours (Ref).

Tooth extraction in patients with hemostatic defects

Tooth extraction in patients with hemostatic defects (eg, hemophilia, von Willebrand disease, other factor deficiencies associated with bleeding) (adjunctive agent):

Note: Generally used in conjunction with (and **not** as a substitute for) replacement of the appropriate clotting factor, especially in individuals with hemophilia. Do not give simultaneously with an activated prothrombin complex concentrate, as this can increase the risk of thromboembolism; if used concurrently, separate by ≥ 12 hours (Ref). Consultation with a hemophilia treatment center is advised.

IV: 10 mg/kg using actual body weight (usual dose range: 500 mg to 1 g) administered ~2 hours before procedure at a rate not to exceed 100 mg/minute (generally over 10 to 20 minutes), then 10 mg/kg 3 to 4 times daily for 2 to 8 days. Alternatively, 10 mg/kg as a single dose ~2 hours prior to procedure; following procedure, transition to oral tranexamic acid depending on individual patient characteristics, type of procedure, other therapies, and degree of bleeding (Ref).

Oral: 25 mg/kg (usual dose range: 1 to 1.5 g) given 2 hours prior to procedure, then 25 mg/kg (usual dose range: 1 to 1.5 g) 3 to 4 times daily for up to 7 to 10 days (Ref).

Trauma-associated hemorrhage or traumatic brain injury

Trauma-associated hemorrhage or traumatic brain injury (off-label use):

Note: Consider for use in patients with significant hemorrhage, at risk of significant hemorrhage, or in moderate traumatic brain injury (TBI) (Glasgow Coma Scale [GCS] score >8 and <13); patients with severe TBI (GCS score 3 to 8) may not demonstrate benefit (Ref).

IV: Loading dose: 1 g over 10 minutes started within 3 hours of injury, followed by 1 g over the next 8 hours as a continuous infusion. **Note:** Some experts suggest using thromboelastogram or rotational thromboelastometry to guide therapy (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Tranexamic acid is $>95\%$ eliminated by the kidney. Dosing recommendations may vary by institution; also consult institutional protocols. No dosage adjustment necessary for indications requiring only 1 to 2 doses (Ref). Tranexamic acid oral rinses have limited absorption (Ref); therefore, the need for renal dose adjustment is unlikely when used for a limited period of time (Ref). There are no data on bioavailability of tranexamic acid administered via nebulization; renal dose adjustment recommendations cannot be provided; use with caution (Ref).

Altered kidney function:

IV:

Intermittent injection (Ref): **Note:** The following adjustments are based on a usual recommended dose of 10 mg/kg or 1 g 3 to 4 times daily.

Serum creatinine <1.4 mg/dL (<120 micromol/L): No dosage adjustment necessary.

Serum creatinine ≥ 1.4 to <2.8 mg/dL (≥ 120 micromol/L to <250 micromol/L): Administer usual dose twice daily.

Serum creatinine ≥ 2.8 to <5.7 mg/dL (≥ 250 to <500 micromol/L): Administer usual dose once daily.

Serum creatinine ≥ 5.7 mg/dL (≥ 500 micromol/L): Administer usual dose every 48 hours **or** 50% of the usual dose once daily.

Continuous infusion:

General recommendations (Ref): **Note:** Developed using the principal that % to be administered is (actual GFR divided by normal GFR) $\times 100\%$, with normal GFR = 90 mL/minute/1.73 m². Recommendations should only be considered for cardiac or spinal indications; not applicable to trauma patients (Ref).

Maintenance infusion following loading dose:

eGFR ≥ 90 mL/minute/1.73 m²: Administer 100% of the usual maintenance rate.

eGFR 60 to <90 mL/minute/1.73 m²: Administer 66% to 100% of the usual maintenance rate.

eGFR ≥ 30 to <60 mL/minute/1.73 m²: Administer 33% to 66% of the usual maintenance rate.

eGFR <30 mL/minute/1.73 m²: Administer 17% to 33% of the usual maintenance rate.

Cardiac surgery regimen-specific examples: **Note:** These are examples of published regimens utilizing a continuous infusion; optimal regimen is uncertain; also refer to institutional protocol.

eGFR-based regimen (BART regimen) (Ref):

eGFR ≥ 90 mL/minute/1.73 m²: Loading dose: 30 mg/kg followed by 16 **mg/kg/hour**.

eGFR 60 to <90 mL/minute/1.73 m²: Loading dose: 30 mg/kg followed by 11 to 16 **mg/kg/hour**.

eGFR >30 to <60 mL/minute/1.73 m²: Loading dose: 25 to 30 mg/kg followed by 5 to 10 **mg/kg/hour**.

eGFR ≤30 mL/minute/1.73 m²: Loading dose: 25 to 30 mg/kg followed by 3 to 5 **mg/kg/hour**.

Serum creatinine-based regimen (Ref):

Serum creatinine 1.6 to 3.3 mg/dL (141 to 292 micromol/L): Reduce maintenance infusion to 1.5 mg/kg/hour (based on a 25% reduction from 2 mg/kg/hour).

Serum creatinine 3.3 to 6.6 mg/dL (292 to 583 micromol/L): Reduce maintenance infusion to 1 mg/kg/hour (based on a 50% reduction from 2 mg/kg/hour).

Serum creatinine >6.6 mg/dL (>583 micromol/L): Reduce maintenance infusion to 0.5 mg/kg/hour (based on a 75% reduction from 2 mg/kg/hour).

Oral (Ref): Note: The following adjustments are based on a usual recommended dose of 10 to 15 mg/kg or 1 to 1.5 g 3 to 4 times daily.

Serum creatinine <1.4 mg/dL (<120 micromol/L): No dosage adjustment necessary.

Serum creatinine ≥1.4 to <2.8 mg/dL (≥120 to <250 micromol/L): Administer usual dose twice daily.

Serum creatinine ≥2.8 to <5.7 mg/dL (≥250 to <500 micromol/L): Administer usual dose once daily.

Serum creatinine ≥5.7 mg/dL (≥500 micromol/L): Administer usual dose every 48 hours, or 50% of the usual dose every 24 hours.

Hemodialysis, intermittent (thrice weekly): Likely to be dialyzable (low protein binding, low V_d (Ref)). **Note:** Because tranexamic acid is dialyzable, schedule intermittent doses after hemodialysis when possible.

IV (intermittent injection), Oral: Administer usual dose every 48 hours, or 50% of the usual dose every 24 hours (Ref).

IV (continuous infusion): Dose as for eGFR <30 mL/minute/1.73 m² (Ref).

Peritoneal dialysis: Likely to be dialyzable (low protein binding, low V_d (Ref)).

IV (intermittent injection), Oral: Administer usual dose every 48 hours, or 50% of the usual dose every 24 hours (Ref).

IV (continuous infusion): Dose as for eGFR <30 mL/minute/1.73 m² (Ref).

CRRT: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent flow rates of 20 to 25 mL/kg/hour (or ~1,500 to 3,000 mL/hour) unless otherwise noted. Close monitoring of response and adverse reactions (eg, seizures, thrombotic events) due to drug accumulation is important.

IV (intermittent injection), Oral: Administer usual dose twice daily (Ref).

IV (continuous infusion): Dose as for eGFR 30 to 60 mL/minute/1.73 m² (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Close monitoring of response and adverse reactions (eg, seizures, thrombotic events) due to drug accumulation is important.

IV (intermittent injection), Oral: Administer usual dose twice daily (Ref).

IV (continuous infusion): Dose as for eGFR 30 to 60 mL/minute/1.73 m² (Ref).

Dosing: Liver Impairment: Adult

No dosage adjustment is necessary.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Tranexamic acid: Pediatric drug information")

Diffuse alveolar hemorrhage, treatment

Diffuse alveolar hemorrhage (intractable), treatment: Very limited data available (Ref), ideal dose-response not established:

Children ≤ 25 kg: Inhaled: 250 mg every 6 hours for 3 to 4 doses (18 to 24 hours); if response occurs, continue treatment for another 2 to 3 doses after bleeding completely stops; if no or minimal response or bleeding worsens, add inhaled recombinant factor VIIa; maximum duration of inhaled therapy: 3 days.

Children >25 kg and Adolescents: Inhaled: 500 mg inhaled every 6 hours for 3 to 4 doses (18 to 24 hours); if response occurs, continue treatment for another 2 to 3 doses after bleeding completely stops; if no or minimal response or bleeding worsens, add inhaled recombinant factor VIIa; maximum duration of inhaled therapy: 3 days.

Dosing based on a prospective pilot study of 18 children (median age: 24 months [interquartile range: 11.3 to 58.5 months]) with intractable diffuse alveolar hemorrhage (DAH) who received inhaled tranexamic acid; DAH responded to inhaled tranexamic treatment alone in 10 children (56%); the 8 nonresponders had inhaled recombinant factor VIIa added; 75% (n=6) of these patients had complete cessation of DAH; none of the patients who responded to treatment had recurrence or complications reported (Ref). Two retrospective studies have reported doses of 250 to 500 mg every 6 to 12 hours until resolution of bleeding for pulmonary hemorrhages (Ref).

Heavy menstrual bleeding

Heavy menstrual bleeding: Postmenarche female: Tablet (Lysteda): Oral: 1,300 mg 3 times daily for up to 5 days during monthly menstruation; maximum daily dose: 3,900 mg/day.

Hereditary angioedema, prophylaxis

Hereditary angioedema (HAE), prophylaxis: Limited data available:

Long-term prophylaxis: **Note:** Not the preferred treatment option; reserve use for when C1-inhibitor concentrate is unavailable (Ref).

Children and Adolescents: Oral: 20 to 50 mg/kg/day in 2 to 3 divided doses; doses up to 75 mg/kg/day have been reported; maximum daily dose range: 3,000 to 6,000 mg/day (Ref); may consider alternate-day regimen or twice-weekly regimen when frequency of attacks reduces (Ref).

Short-term prophylaxis (eg, prior to surgical or diagnostic interventions in head/neck region): **Note:** Not the preferred treatment option; some experts do not recommend use for short-term prophylaxis (Ref).

Weight directed: Children and Adolescents: Oral: 20 to 50 mg/kg/day in 2 to 3 divided doses; maximum daily dose range: 3,000 to 6,000 mg/day; initiate therapy at least 5 days before and continue for 2 days postprocedure (Ref).

Fixed dosing: Children and Adolescents: Patients with an adequate weight (eg, ≥ 50 kg): Oral: 500 mg 4 times daily (Ref); therapy usually initiated 2 to 5 days before dental work and continue for 2 days after the procedure (Ref).

Prevention of bleeding associated with tooth extraction in hemophilic patients

Prevention of bleeding associated with tooth extraction in hemophilic patients:

Note: Use in combination with replacement therapy.

Infants, Children, and Adolescents: IV: 10 mg/kg immediately before surgery, then 10 mg/kg/dose 3 to 4 times daily for 2 to 8 days.

Prevention of perioperative bleeding

Prevention of perioperative bleeding: Limited data available; reported regimens variable and ideal dose-response not established:

General dosing (non-cardiac): Infants, Children, and Adolescents: IV: Loading dose: 10 to 30 mg/kg followed by a continuous IV infusion at 5 to 10 mg/kg/hour; dosing based on a pharmacokinetic model to achieve a target serum concentration of 20 mcg/mL and 70 mcg/mL, respectively (Ref).

Cardiac surgery with cardiopulmonary bypass: Infants, Children, and Adolescents:

Low dose: IV: Loading dose: 10 mg/kg followed by a continuous IV infusion at 5 mg/kg/hour; dosing based on a pharmacokinetic model to achieve a target serum concentration of 20 mcg/mL; tranexamic acid must also be added to cardiopulmonary bypass solution at a concentration of 20 mcg/mL (Ref). A pharmacokinetic analysis (n=43; mean age: 123 days [range: 6 to 348 days]; mean weight: 4.95 kg [range: 2.3 to 9.5 kg]) targeting a serum concentration of 20 mcg/mL proposed the following regimen: Loading dose: 10 mg/kg, followed by a continuous IV infusion at 10 mg/kg/hour until initiation of cardiopulmonary bypass, then IV priming bolus: 4 mg/kg into the bypass prime volume, followed by a continuous IV infusion at 4 mg/kg/hour (Ref). Another regimen studied in 2 trials (n=80; age range: 2 months to 15 years) is 10 mg/kg into the bypass circuit after induction, during cardiopulmonary bypass, and after protamine reversal of heparin for a total of 3 doses (Ref). A pharmacokinetic analysis has proposed the following regimen to achieve a target serum concentration range of 20 to 30 mcg/mL in children 1 to 12 years and weighing 5 to 40 kg: IV: Loading dose: 6.4 mg/kg over 5 minutes followed by a weight-adjusted continuous IV infusion in the range of 2 to 3.1 mg/kg/hour; the pharmacokinetic data showed that patients weighing less should receive an initial continuous IV infusion rate at the higher end of the range (ie, if patient weight=5 kg then initial continuous IV infusion rate: 3.1 mg/kg/hour; if patient weight=40 kg then initial continuous IV infusion rate: 2 mg/kg/hour) (Ref).

Intermediate dose: IV: Loading dose: 30 mg/kg followed by a continuous IV infusion at 10 mg/kg/hour; dosing based on a pharmacokinetic model to achieve a target serum concentration of 70 mcg/mL; tranexamic acid must also be added to cardiopulmonary bypass solution at a concentration of 70 mcg/mL (Ref).

High dose: IV: Loading dose: 50 mg/kg, followed by a continuous IV infusion at 15 mg/kg/hour and 50 mg/kg priming dose into the circuit when bypass initiated (Ref); dosing based on a pharmacokinetic model to achieve a target serum concentration of 150 mcg/mL (Ref).

Spinal surgery (eg, idiopathic scoliosis): Children ≥ 8 years and Adolescents: IV: Loading dose: 100 mg/kg, followed by a continuous IV infusion at 10 mg/kg/hour until skin closure (Ref). Other reported regimens with positive results: Loading dose: 20 mg/kg, followed by a continuous IV infusion at 10 mg/kg/hour (Ref); loading dose: 10 mg/kg, followed by a continuous IV infusion at 1 mg/kg/hour (Ref); loading dose: 50 mg/kg, followed by a continuous IV infusion at 5 mg/kg/hour (Ref).

Craniosyntosis surgery: Infants ≥ 2 months and Children ≤ 6 years: IV: Loading dose: 50 mg/kg over 15 minutes prior to incision, followed by a continuous IV infusion at 5 mg/kg/hour until skin closure (Ref) **or** loading dose: 15 mg/kg over 15 minutes prior to incision, followed by a continuous IV infusion at 10 mg/kg/hour until skin closure (Ref). Other reported regimens with positive results: Loading dose: 10 mg/kg at start of surgery, followed by a continuous IV infusion at 5 mg/kg/hour for 24 hours postoperatively (Ref).

Trauma, hemorrhagic

Trauma, hemorrhagic (acute traumatic coagulopathy): Limited data available:

Note: Reported regimens are variable and ideal dose-response is not established:

Children < 12 years: IV: Loading dose: 15 mg/kg over 10 minutes given within 3 hours of injury (maximum dose: 1,000 mg/dose), followed by continuous IV infusion at 2 mg/kg/hour for ≥ 8 hours or until bleeding stops (Ref).

Children ≥ 12 years and Adolescents: IV: Loading dose: 1,000 mg over 10 minutes given within 3 hours of injury, followed by 1,000 mg infused over 8 hours (Ref).

Traumatic hyphema

Traumatic hyphema: Limited data available: Children and Adolescents: Oral: 25 mg/kg/dose every 8 hours for 5 to 7 days (Ref). **Note:** This same regimen may also be used for secondary hemorrhage after an initial traumatic hyphema event.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Note: Recommendations are dependent on use and route.

Oral:

Menorrhagia: Female Children ≥ 12 years and Adolescents:

$S_{cr} > 1.4$ to ≤ 2.8 mg/dL: 1,300 mg twice daily for up to 5 days during monthly menstruation.

$S_{cr} > 2.8$ to ≤ 5.7 mg/dL: 1,300 mg once daily for up to 5 days during monthly menstruation.

$S_{cr} > 5.7$ mg/dL: 650 mg once daily for up to 5 days during monthly menstruation.

Prophylaxis of hereditary angioedema: Children and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling; however, due to risk of accumulation with kidney impairment, dosage adjustments are recommended (Ref).

IV:

Tooth extraction in patients with hemophilia: Infants, Children, and Adolescents:

$S_{cr} 1.36$ to ≤ 2.83 mg/dL: Maintenance dose of 10 mg/kg/dose twice daily.

$S_{cr} > 2.83$ to ≤ 5.66 mg/dL: Maintenance dose of 10 mg/kg/dose once daily.

$S_{cr} > 5.66$ mg/dL: Maintenance dose of 10 mg/kg/dose every 48 hours **or** 5 mg/kg/dose every 24 hours.

Prophylaxis or treatment of mild to major bleeding secondary to trauma or surgery : Infants, Children, and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling; however, due to risk of accumulation with kidney impairment, dosage adjustments are recommended (Ref).

Dosing: Liver Impairment: Pediatric

Infants, Children, and Adolescents: No adjustment is necessary.

Adverse Reactions (Significant): Considerations

Hypersensitivity reactions (immediate and delayed)

Immediate hypersensitivity reactions (eg, **anaphylaxis**) have been reported, ranging from mild **pruritus** and/or **urticaria** to more severe reactions, including **angioedema**, **wheezing**, **hypotension**, and **anaphylactic shock** (Ref). Delayed hypersensitivity reactions include **fixed drug eruption** and **toxic epidermal necrolysis** (TEN) (Ref).

Mechanism: Immediate hypersensitivity reactions (eg, anaphylaxis, urticaria): Non-dose-related, immunologic, IgE-mediated. Delayed hypersensitivity reactions: Non-dose-related, immunologic, T-cell mediated (Ref).

Onset: Immediate hypersensitivity reactions: Rapid; occur within 1 hour of administration but may occur up to 6 hours after exposure (Ref). Delayed hypersensitivity reactions: Varied; serious cutaneous adverse reactions, including TEN, occur 1 to 8 weeks after initiation (Ref).

Ocular effects

Visual defects (eg, **vision color changes**, **visual impairment**, **vision loss**), **retinal artery occlusion**, and **retinal venous occlusion** have been reported (Ref). **Conjunctivitis** (ligneous), a form of membranous conjunctivitis, has been reported with the oral formulation but resolved upon discontinuation of therapy (Ref).

Mechanism: Visual defects: Unknown (Ref). Ligneous conjunctivitis: Unknown; likely vasculopathy with increased vessel permeability and consequent loss of protein (Ref).

Onset: Visual defects: Varied; retinal artery occlusion occurred 5 days to 1 month after initiation (Ref). Ligneous conjunctivitis: Varied; occurred 1 to 9 months after initiation; upon reintroduction, occurred within 2 days (Ref).

Risk factors:

- Kidney failure (Ref)

Seizures and myoclonus

Seizures have been reported, typically generalized tonic-clonic; although, focal and mixed seizures may also occur (Ref). **Myoclonus** is noted in ~20% of patients. Seizures usually persist for a few minutes and rarely progress to status epilepticus (Ref). Recurrent seizures may occur in 30% to 60% of patients during the first 24 to 48 hours after postoperative administration (Ref). The incidence of seizures is ~2.7% (Ref).

Mechanism: Dose-related: Unknown; may occur due to antagonistic effect of tranexamic acid at GABA-A receptors and neural glycine receptors, resulting in lowered seizure threshold; cerebral emboli may also play a role (Ref).

Onset: Rapid; within the first 5 to 8 hours after postoperative administration (Ref).

Risk factors:

- Higher intravenous doses may increase risk (Ref); although, one study showed no increased risk of seizure with high dose versus low dose (Ref)
- Cardiac surgery, in particular patients undergoing open heart surgery; deep hypothermic circulatory arrest, increased cardiopulmonary bypass time, or prolonged aortic cross-clamp time are also associated with increased risk in this population (Ref)
- Females (Ref)
- Increased age (>70 years) (Ref)
- Poor overall health (Ref)
- Kidney impairment (Ref)
- Prior neurological and cardiovascular disorders (Ref)
- History of stroke (Ref)

Thromboembolic events

Venous thromboembolism and **arterial thromboembolism**, including **deep vein thrombosis**, **pulmonary embolism**, **retinal vein occlusion**, and **retinal artery occlusion**, have been reported (Ref). The incidence of thromboembolism after various surgical procedures has been reported from 0.3% to 8.2% (Ref). In contrast, some studies have shown no significant increase in thromboembolism risk (Ref).

Mechanism: Dose-related; unknown; reversibly displaces plasminogen from fibrin, resulting in the cessation of fibrinolysis (Ref). May also inhibit the proteolytic activity of plasmin (Ref).

Onset: Varied; may occur within 1 hour (Ref), up to several years after initiation (Ref).

Risk factors:

- High doses (≥ 20 mg/kg) or 2 to 4 g (Ref)
- Variceal bleeding or liver disease (Ref)
- Increased age (Ref)
- Cardiovascular disease (Ref)

- History of or active thromboembolic disease
- Concurrent procoagulant agents (eg, prothrombin complex concentrate, oral tretinoin, hormonal contraceptives)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. As reported with oral formulation unless otherwise noted.

>10%:

Gastrointestinal: Abdominal pain (20%)

Nervous system: Headache (50%)

Neuromuscular & skeletal: Back pain (21%), musculoskeletal pain (11%)

Respiratory: Nasal signs and symptoms (25%; including sinus symptoms)

1% to 10%:

Hematologic & oncologic: Anemia (6%)

Nervous system: Fatigue (5%)

Neuromuscular & skeletal: Arthralgia (7%), muscle cramps ($\leq 7\%$), muscle spasm ($\leq 7\%$)

Postmarketing (all formulations):

Cardiovascular: Arterial thromboembolism (Meaidi 2021), deep vein thrombosis (HALT-IT 2020), hypotension (with rapid IV injection), pulmonary embolism (HALT-IT 2020; Ijaopo 2020), venous thromboembolism (HALT-IT 2020)

Dermatologic: Allergic dermatitis (Imbesi 2010), fixed drug eruption (Kaku 2014), pruritus (Imbesi 2010), toxic epidermal necrolysis (Pretel Irazabal 2013), urticaria (Imbesi 2010)

Gastrointestinal: Diarrhea, nausea, vomiting

Hypersensitivity: Anaphylactic shock, anaphylaxis (El Hanache 2021), angioedema (Imbesi 2010), nonimmune anaphylaxis

Nervous system: Cerebral thrombosis, dizziness, myoclonus (Lecker 2016), seizure (Lecker 2016)

Ophthalmic: Chromatopsia, conjunctivitis (ligneous) (Song 2014), retinal artery occlusion (Wijetilleka 2017), retinal vein occlusion, vision color changes (Kiser 2021), vision loss (Wijetilleka 2017), visual impairment (Kitamura 2003)

Renal: Renal cortical necrosis (Ko 2017)

Respiratory: Wheezing (Murdaca 2020)

Contraindications

Hypersensitivity to tranexamic acid or any component of the formulation.

Injection: Active intravascular clotting; subarachnoid hemorrhage; neuraxial (ie, intrathecal, epidural) injection.

Oral: Active thromboembolic disease (eg, cerebral thrombosis, DVT, or pulmonary embolism); history of thrombosis or thromboembolism, including retinal vein or retinal artery occlusion; intrinsic risk of thrombosis or thromboembolism (eg, hypercoagulopathy, thrombogenic cardiac rhythm disease, thrombogenic valvular disease); patients using combined hormonal contraception who may become pregnant.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindications (not in the US labeling): Injection, oral: History or risk of thrombosis (unless concurrent anticoagulation therapy is possible); hematuria.

Warnings/Precautions

Concerns related to adverse effects:

- CNS effects: May cause dizziness, which may impair physical or mental abilities; patients must be cautioned about performing tasks which require mental alertness (eg, operating machinery or driving).

Disease-related concerns:

- Disseminated intravascular coagulation: Use with extreme caution in patients with disseminated intravascular coagulation requiring antifibrinolytic therapy; patients should be under strict supervision of a health care provider experienced in treating this disorder.
- Renal impairment: Use with caution in patients with renal impairment; dosage modification necessary.
- Subarachnoid hemorrhage: Use with caution in patients with subarachnoid hemorrhage; cerebral edema and infarction may occur.
- Vascular disease: Use with caution in patients with uncorrected cardiovascular or cerebrovascular disease due to the complications of thrombosis.

Dosage form specific issues:

- Injection: For IV use only. Label syringes clearly with the IV route of administration; confirm the correct route of administration to avoid confusion with other injectable solutions that may be administered at the same time.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous:

Generic: 1000 mg/10 mL (10 mL)

Solution, Intravenous [preservative free]:

Cyklokapron: 1000 mg/10 mL (10 mL)

Generic: 1000 mg/10 mL (10 mL); 1000 mg/100 mL in NaCl 0.7% (100 mL)

Tablet, Oral:

Lysteda: 650 mg [DSC]

Generic: 650 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Cyklokapron Intravenous)

1000 mg/10 mL (per mL): \$2.40

Solution (Tranexamic Acid Intravenous)

1000 mg/10 mL (per mL): \$0.19 - \$8.68

Solution (Tranexamic Acid-NaCl Intravenous)

1000MG/100ML 0.7% (per mL): \$0.08 - \$0.25

Tablets (Tranexamic Acid Oral)

650 mg (per each): \$5.21 - \$5.22

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Cyklokapron: 100 mg/mL (5 mL, 10 mL)

Generic: 100 mg/mL (5 mL, 10 mL, 50 mL); 1000 mg/10 mL (10 mL)

Tablet, Oral:

Cyklokapron: 500 mg

Generic: 500 mg

Extemporaneous Preparations

50 mg/mL (5%) Oral Solution

A 50 mg/mL (5%) oral solution may be prepared by diluting 5 mL of 10% (100 mg/mL) tranexamic acid injection with 5 mL sterile water. Label "refrigerate". Stable for 5 days refrigerated (Ref).

25 mg/mL Oral Suspension

A 25 mg/mL oral suspension may be prepared with tablets. Place one 500 mg tablet (strength not available in US) into 20 mL water and let stand ~2 to 5 minutes. Begin stirring and continue until the tablet is completely disintegrated, forming a fine particulate suspension (dispersion time for each 500 mg tablet is ~2 to 5 minutes). Administer immediately after preparation (Ref).

Administration: Adult

Inhalation via nebulization (off-label use/route): Administer over 15 minutes via jet nebulizer (Ref).

IV:

Note: For IV administration only. Refer to institutional policies and procedures. For indication-specific administration via continuous infusion, refer to adult dosing field for detailed recommendations.

Undiluted solution (100 mg/mL): Administer 0.5 **mL/minute** (50 **mg/minute**); to avoid hypotension do not exceed 1 **mL/minute** (100 **mg/minute**).

Diluted solution (10 mg/mL): Administer 5 **mL/minute** (50 **mg/minute**); to avoid hypotension do not exceed 10 **mL/minute** (100 **mg/minute**).

Diluted solution (20 mg/mL): Administer 2.5 **mL/minute** (50 **mg/minute**); to avoid hypotension do not exceed 5 **mL/minute** (100 **mg/minute**).

Oral: Administer without regard to meals. Swallow tablet whole; do not break, chew, or crush.

Preparation for Administration: Adult

Tranexamic acid does not require dilution; however, doses using ampules/vials may be diluted in 50 to 250 mL of LR, NS, or D5W to a final concentration of 10 mg/mL or 20 mg/mL (Ref). Gently invert

infusion bag; do not shake. Discard any unused solution; refer to institutional protocols. **Note:** A 10 mg/mL premixed solution formulation is also available.

Administration: Pediatric

Oral: Administer without regard to meals; tablets should be swallowed whole; do not break, split, chew, or crush.

Parenteral:

Intermittent IV dose: May be administered undiluted by direct IV injection at a maximum rate of 100 mg/minute; faster rates may cause hypotension.

Continuous IV infusion:

Loading dose: May be administered either undiluted or diluted in a compatible diluent; infuse over 5 to 15 minutes (Ref). Neonatal patients received loading doses over 60 minutes (Ref).

IV infusion: Following dilution, administer by continuous IV infusion at a rate not to exceed 100 mg/minute.

Inhalation: Administer undiluted (100 mg/mL) by jet nebulization (Ref); time for nebulization average 15 minutes in adults (Ref).

Preparation for Administration: Pediatric

Parenteral: Continuous IV infusion:

Loading dose: May be further diluted in 1 mL/kg or 20 mL NS (Ref).

IV infusion: According to the manufacturer, tranexamic acid may be mixed with most solutions for infusion such as electrolyte solutions, carbohydrate solutions, amino acid solutions, and dextran solutions. Based on adult studies, ampules/vials may be diluted in 50 to 250 mL of NS or D5W (Ref). ASHP recommends a standard concentration of 100 mg/mL (undiluted drug) (Ref). **Note:** A 10 mg/mL premixed solution formulation is also available.

Storage/Stability

IV:

Ampules and vials: Store intact ampules and vials at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Diluted mixture may be stored for up to 4 hours at room temperature. However, solutions prepared in NS are chemically stable for up to 180 days at room temperature (McCluskey 2014). In another study, tranexamic acid (undiluted) was shown to be chemically stable for up to 12 weeks when stored at -20°C, 4°C, 22°C, and 50°C; freezing tranexamic acid in original ampules is unacceptable due to cracking of the ampules (de Guzman 2013). Freezing tranexamic acid in original vials has not been evaluated.

Premixed solution: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F).

Tablets: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Abnormal uterine bleeding, nonacute (oral): Treatment of cyclic heavy menstrual bleeding.

Limitations of use: For use in patients who may become pregnant; not for use prior to menarche or post menopause.

Tooth extraction in patients with hemostatic defects (injection, oral [Cyklokapron; Canadian product]): Short-term use in hemophilia patients to reduce or prevent hemorrhage and reduce need for replacement therapy during and following tooth extraction.

Use: Off-Label: Adult

Abnormal uterine bleeding, acute; Dental procedures in patients on oral anticoagulant therapy; Hemoptysis (nonmassive), treatment; Hereditary angioedema, long-term prophylaxis; Hereditary hemorrhagic telangiectasia, epistaxis or other bleeding sites; Intracranial hemorrhage associated

with thrombolytic treatment; Perioperative prevention of blood loss and transfusion, cardiac surgery; Perioperative prevention of blood loss and transfusion, orthopedic surgery (hip or knee arthroplasty); Perioperative prevention of blood loss and transfusion, spinal surgery; Postpartum hemorrhage, prevention; Postpartum hemorrhage, treatment; Subarachnoid hemorrhage, prevention of early aneurysmal rebleeding; Trauma-associated hemorrhage or traumatic brain injury

Medication Safety Issues

Sound-alike/look-alike issues:

Cyklokapron may be confused with cycloSPORINE

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication (injectable formulation) among its list of drugs that have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Administration issues:

Inadvertent administration of tranexamic acid by the epidural or spinal route during neuraxial (eg, epidural, spinal) anesthesia has led to potentially fatal neurotoxic adverse reactions. Carefully evaluate storage procedures within the surgical suite, including separating tranexamic acid from local anesthetics; consider additional prevention measures, including purchasing, dispensing, and administration (ISMP [Smetzer] 2019; NAN Alert 2020; Patel 2019).

Other safety concerns:

TXA is an error-prone abbreviation (mistaken as TPA [tissue plasminogen activator])

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Anti-inhibitor Coagulant Complex (Human): Antifibrinolytic Agents may increase thrombogenic effects of Anti-inhibitor Coagulant Complex (Human). *Risk X: Avoid*

Estrogen Derivatives: May increase thrombogenic effects of Tranexamic Acid. *Risk X: Avoid*

Factor IX Complex (Human) [(Factors II, IX, X)]: Antifibrinolytic Agents may increase adverse/toxic effects of Factor IX Complex (Human) [(Factors II, IX, X)]. Specifically, the risk for thrombosis may be increased. *Risk X: Avoid*

Hormonal Contraceptives: May increase thrombogenic effects of Tranexamic Acid. *Risk X: Avoid*

Prothrombin Complex Concentrate (Human) [(Factors II, VII, IX, X), Protein C, and Protein S]: Antifibrinolytic Agents may increase adverse/toxic effects of Prothrombin Complex Concentrate (Human) [(Factors II, VII, IX, X), Protein C, and Protein S]. Specifically, the risk for thrombosis may be increased. *Risk X: Avoid*

Thrombolytic Agents: Tranexamic Acid may decrease therapeutic effects of Thrombolytic Agents. Thrombolytic Agents may decrease therapeutic effects of Tranexamic Acid. *Risk X: Avoid*

Tretinoin (Systemic): May increase thrombogenic effects of Antifibrinolytic Agents. Management: Concomitant use of antifibrinolytics and tretinoin is not recommended. If combined, monitor patients closely for any signs of thrombotic complications. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Tranexamic acid is an alternative agent for the treatment of heavy menstrual bleeding and one option for patients who desire future fertility (ACOG 2019). The manufacturer recommends non-hormonal contraception during treatment, as hormonal contraceptives may increase the risk of thromboembolic events (use of hormonal contraception is contraindicated by some manufacturers). However, tranexamic acid in combination with oral contraceptives may be considered for the treatment of heavy menstrual bleeding when monotherapy is ineffective and other treatment options have failed (ACOG 2013; ACOG 2019).

Pregnancy Considerations

Tranexamic acid crosses the placenta; concentrations within cord blood are similar to maternal serum.

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of oral tranexamic acid may be altered (Muhunthan 2020).

Oral tranexamic acid is used off label for the long-term prophylaxis of hereditary angioedema (HAE) and use for this indication in pregnant patients has been reported (González-Quevedo 2016; Loli-Ausejo 2020; Machado 2017; Milingos 2009). Tranexamic acid is not the preferred therapy for HAE in pregnant patients. Use may be considered for long-term prophylaxis of HAE during pregnancy when preferred treatment is not available; however, efficacy data are not available. Patients with HAE should be monitored closely during pregnancy and for at least 72 hours after delivery (US HAEA [Busse 2021]; WAO/EAACI [Maurer 2022]).

IV tranexamic acid is used off label for the treatment of postpartum hemorrhage (PPH) (Ducloy-Bouthors 2011; RCOG [Pavord 2017]; WOMAN Trial Collaborators 2017). A significant reduction in risk of death due to bleeding was observed when treatment was started within 3 hours of vaginal birth or cesarean section (WOMAN Trial Collaborators 2017). Tranexamic acid is recommended for the treatment of obstetric hemorrhage when initial medical management fails (ACOG 2017; WHO 2017).

IV tranexamic acid has also been studied for prophylaxis of PPH in patients prior to vaginal or cesarean delivery (Novikova 2015; Saccone 2019; Sentilhes 2018; Sentilhes 2021; Simonazzi 2016; Xia 2020). Tranexamic acid may be considered as adjunctive therapy in patients at high risk for PPH. Patients at high risk include those with a known coagulation defect; bleeding upon admission; hematocrit <30%; history of PPH; abnormal vital signs (hypotension and tachycardia); or placenta previa, accreta, increta, or percreta. However, available data related to prophylactic use are insufficient, and use for routine prophylaxis against PPH is not currently recommended outside of the context of clinical research (ACOG 2017; Muñoz 2019). Additional studies may be needed to define the optimal dose (Ahmadzia 2021; Li 2021).

When managing PPH, do not use tranexamic acid in patients with clear contraindications to therapy, including active intravascular clotting, known thromboembolic event during pregnancy, known hemostatic disorders, or known hypersensitivity to tranexamic acid (Ducloy-Bouthors 2011; Shakur 2010; WHO 2017).

Breastfeeding Considerations

Tranexamic acid is present in breast milk.

Breast milk concentrations of tranexamic acid in patients who were lactating were ~1% of the maximum maternal serum concentration when measured 1 hour after the last dose following 2 days of treatment (maternal dose and actual milk concentrations not provided) (Verstraete 1985).

Thromboembolic disorders were not observed in breastfed infants following maternal use of tranexamic acid for the treatment of postpartum hemorrhage (WOMAN Trial Collaborators 2017). An increased risk of adverse events was not observed in 21 breastfed infants exposed to tranexamic acid following maternal use for coagulation disorders (maternal dose range: 1.5 to 4 g/day). Authors of this study suggest taking the maternal dose immediately after breastfeeding to minimize infant exposure and monitor the infant for adverse events (Gilad 2014).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Lactation may increase the frequency of hereditary angioedema (HAE) attacks; patients with HAE should be monitored closely. Tranexamic acid is not the preferred treatment for hereditary angioedema in patients who are breastfeeding (US HAEA [Busse 2021]; WAO/EAACI [Maurer 2022]).

Monitoring Parameters

Ophthalmic examination (visual acuity, optical coherence tomography) at regular intervals if on long-term therapy (>3 months); signs/symptoms of hypersensitivity reactions, seizures (consider EEG monitoring for patients with history of seizures or who experience myoclonic movements, twitching, or evidence of focal seizures), and thrombotic events; in patients with trauma-associated hemorrhage, thromboelastography (TEG), or rotational thromboelastometry (ROTEM) where available (Colwell 2021).

Mechanism of Action

Forms a reversible complex that displaces plasminogen from fibrin resulting in inhibition of fibrinolysis; it also inhibits the proteolytic activity of plasmin

With reduction in plasmin activity, tranexamic acid also reduces activation of complement and consumption of C1 esterase inhibitor (C1-INH), thereby decreasing inflammation associated with hereditary angioedema.

Pharmacokinetics (Adult Data Unless Noted)

Distribution: V_d : IV: 9 to 12 L; cerebrospinal fluid and aqueous humor of eye concentrations are 10% of plasma.

Protein binding: ~3%, primarily to plasminogen.

Bioavailability: Oral: ~45%.

Half-life elimination: IV: ~2 hours; Oral: ~11 hours.

Time to peak: Oral:

Single dose: Mean: 2.5 hours (range: 1 to 5 hours).

Multiple dose: Mean: 2.5 hours (range: 2 to 3.5 hours).

Excretion: Urine (>95% as unchanged drug).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Following administration of a single IV injection, urinary excretion declines as renal function decreases.

Pediatric: The C_{max} and AUC values after a single oral dose of 1,300 mg in adolescent females were 20% to 25% less than those in adult females given the same dose.

In vitro data suggests that neonates require a lower serum tranexamic acid concentration than adults (6.54 mcg/mL vs 17.5 mcg/mL) to completely prevent fibrinolysis (Yee 2013). In pediatric patients weighing 5 to 40 kg undergoing cardiac surgery with by-pass, a target serum concentration range of 20 to 30 mcg/mL has been used in pharmacokinetic analysis (Dowd 2002; Grassin-Delyle 2013).

Patient Counseling Points

(For additional information see "Tranexamic acid: Patient drug information")

What is this drug used for?

- It is used to treat or prevent bleeding in patients with hemophilia after having a tooth removed.
- It is used to treat heavy bleeding during monthly periods (menstruation).
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Tablets:

- Headache
- Stomach pain
- Back, muscle, or joint pain
- Nose stuffiness
- Muscle cramps
- Feeling tired or weak

Injection:

- Diarrhea, upset stomach, or throwing up
- Dizziness

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Flushing
- Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight
- Change in eyesight, eye pain, or severe eye irritation
- Seizures
- Twitching
- Trouble controlling body movements
- Blood clot like chest pain or pressure; coughing up blood; shortness of breath; swelling, warmth, numbness, change of color, or pain in a leg or arm; or trouble speaking or swallowing
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

- Severe dizziness or passing out

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits

of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Cyklokapron | Exacyl | Kapron | Tafixyl;
- (AR) Argentina: Arotran | Lysteda | Tranexal;
- (AT) Austria: Cyklokapron | Tranexamsaeure accord | Tranexamsaeure medicamentum | Tranexamsaeure medicamentum;
- (AU) Australia: Apo tranexamic acid | Cyklokapron | Tranexamic acid juno | Tranexamic acid link | Tranexamic acid lupin | Tranexamic acid myx | Tranexamic aft | Zamic;
- (BD) Bangladesh: Fibrino | Hemostat | Hemotrax | Tranexil | Transamin | Trexam | Xamic;
- (BE) Belgium: Exacyl;
- (BF) Burkina Faso: Trenaxa;
- (BG) Bulgaria: Exacyl | Medocapron;
- (BR) Brazil: Acido tranexamico | Hemoblock | Traneger | Transamin | Trexacont;
- (CH) Switzerland: Cyklokapron | Tranexam orpha | Tranexamic orpha;
- (CI) Côte d'Ivoire: Tranz | Trenaxa;
- (CL) Chile: Ciclokapron | Espercil | Pause | Tranexton | Trexanic;
- (CN) China: Bei rui ning | Dontilyse | Fu kang | Ji xue ning | Jiening | Long yue | Su ning | Ta jiu shu | Tranexamic acid and sodium chloride | Transamic acid | Transamin | Wei xue | Xue bo | Xue li ting | Xue zi ling;
- (CO) Colombia: Acido tranexamico | Clintranex | Cyklokapron | Fibrilok | Fibrinexam | Gotronal | Tarexa | Tranexam | Zociled;
- (CR) Costa Rica: Acido tranexamico;
- (CZ) Czech Republic: Exacyl | Tranexamic acid accord;
- (DE) Germany: Cyklokapron | Tranexamsaeure accord | Tranexamsaeure carino | Tranexamsaeure eberth | Tranexamsaeure hexal | Tranexamsaeure tillomed | Tranexamsaeure tillomed | Ugurol;
- (DO) Dominican Republic: Amchafibrin | Pause;
- (EC) Ecuador: Acido tranexamico | Bonaplasmin | Clintranex | Fibrilok | Gotronal | Hemoblock | Rixam | Tranexam;
- (EE) Estonia: Amchafibrin | Caprilon | Cyclokapron | Cyklokapron | Exacyl | Medsamic | Tranexamic | Tranexamsaeure eberth | Tranexamsaeure tillomed;
- (EG) Egypt: Cyklokapron | Hemokapron | Hemoxamine | Kapron | Savibleed | Taroxatron | Tranex | Trexam;
- (ES) Spain: Acido tranexamico Tillomed | Amchafibrin | Medsamic;
- (ET) Ethiopia: Medsamic;
- (FI) Finland: Caprilon | Cyklokapron | Cyklokapron pfizer | Tranexamic acid accord | Tranexamic acid Alternova | Tranexamic acid baxter | Tranexamic acid orion | Tranexamic acid stragen;
- (FR) France: Acide tranexamique aguettant | Acide tranexamique arrow | Exacyl | Spotof | Tranexamic acid accord;
- (GB) United Kingdom: Cyklo-f | Cyklokapron | Femstrual | Menstralite | Tranexamic | Tranexamic acid kent pharm | Tranexamic acid tillomed;
- (GT) Guatemala: Agetax;
- (HK) Hong Kong: Cyklokapron | Gemotran | Transamin;
- (ID) Indonesia: Asam Traneksamat | Asamnex | Clonex | Cyklokapron | Ditrane | Kalnex | Lunex | Nexitra | Plasminex | Pytramic | Ronex | Tramix | Tranec | Tranexid | Transamin;
- (IE) Ireland: Cyklokapron;
- (IL) Israel: Hexakapron;
- (IN) India: Age gard od | Alflow | Altran | Capitrax | Celtranz | Clip | Clot xl | Clotawin-t | Coastat | Cyklokapron | Desqual | Dubatran | Extam | Glantrax | Glotran | Gynae pil | Melano tx | Menogia |

- Naxamic | Nexamic | Pause | Proxemic sr | Pynomax od | Pynomax tx | Snoglo t | Stalia | Stopbleed | Synostat | Systra | T Stat | T-klot | T-syl | Tanmic | Taxi | Texakind | Texid | Traklot | Tranarest | Trance | Tranecid | Tranee | Tranemic | Tranexid | Tranfib | Tranlok | Tranomac | Tranostat | Transamine | Traptic | Traxage | Traxamic | Traxido | Trenaxa | Triolite | Trxamic | Tyrocin | Wistran | Xamic;
- (IQ) Iraq: Tranexam;
 - (IT) Italy: Acido tranexamico Tillomed | Ugurol;
 - (JO) Jordan: Cyklokapron | Xenamex;
 - (JP) Japan: Hexapromin | Keisamine | Mendamin | Nicolda | Pletasmin | Spiramin | Toranechimu | Toranechimu choseido | Tracapmin | Tranexamic acid teikoku medix | Tranexamic acid | Tranexan taiyo | Transamin | Trasamlon hexal | Youxamine;
 - (KE) Kenya: Menozip | Pause | Synostat | Tafixyl | Texakind | Tranamic | Trantum | Trenaxa | Xenamex;
 - (KR) Korea, Republic of: Cyklokapron;
 - (KW) Kuwait: Cyclokapron | Cyklokapron | Exacyl | Tramific | Tranexamic acid;
 - (LB) Lebanon: Exacyl;
 - (LT) Lithuania: Amchafibrin | Cyklokapron | Cyklokapron axicorp | Exacyl | Tranexaheal | Tranexamsaure eberth;
 - (LU) Luxembourg: Exacyl;
 - (LV) Latvia: Cyclokapron | Cyklokapron | Exacyl;
 - (MA) Morocco: Exacyl;
 - (MX) Mexico: Agetax | Lysteda | Oniren | Solidtrax;
 - (MY) Malaysia: Cyklokapron | Tren;
 - (NG) Nigeria: Cyklatran | Emxamic | Unotation;
 - (NL) Netherlands: Cyklokapron | Tranexaminezuur | Tranexaminezuur Mylan | Traxidot;
 - (NO) Norway: Cyklokapron | Cyklonova | Pilexam | Tracafour | Tranexamic acid stragen;
 - (NZ) New Zealand: Cyklokapron | Tranexamic Acid AFT;
 - (PA) Panama: Acido tranexamico | Acido tranexamico bioindustria l.i.m. | Rixam | Tafixyl | Tranamil | Tranemic;
 - (PE) Peru: Acido tranexam | Acido tranexamico | Acitransesa | Bonaplasmin | Clerilax | Diphaxamico | Exofast | Pause | Plasminex | Rixam | Tramex B | Tranexol | Tranexpalm | Transamin | Traxx;
 - (PH) Philippines: Aneptil | Cycloxiid | Cyklokapron | Dostan | Fimoplas | Nexavell | Proklot | Tarnex | Tranexaprime | Transtat | Trifex | Xanfib | Zantrax;
 - (PK) Pakistan: Dravix | Exirol | Haematrix | Haemic | Hemostat | Hemsamic | Raytrax | Tranol | Transcam | Traumax | Xed;
 - (PL) Poland: Acide tranexamique aguettant | Celtranz | Exacyl | Gemotran | Herajit | Kwas traneksamowy tillomed | Tranemic | Tranexamic acid accord;
 - (PR) Puerto Rico: Cyklokapron | Lysteda;
 - (PT) Portugal: Acido tranexamico accord | Acido tranexamico zentiva | Exacyl | Tafixyl;
 - (QA) Qatar: Cyklokapron | Herajit | Kapron | Tramific | Tranemic | Tranexaheal | Traxal | Xenamex;
 - (RO) Romania: Acido tranexamico | Exacyl | Ugurol;
 - (RU) Russian Federation: Bitranex | Cyclogemal | Cyclohemal | Cyclokapron | Exacyl | Pauz em | Sanxamic | Sanxamik | Stagemin | Teovexal | Traksara | Trameston | Tranavex | Tranexam | Tranexamic acid Akrihin | Tranexamic acid solopharm | Traxara | Troxaminat;
 - (SA) Saudi Arabia: Azeptil | Cyklokapron | Kapron | Lysteda | Tafixyl;
 - (SE) Sweden: Cyklo-f | Cyklokapron | Cyklonova | Statraxon | Tranexa | Tranexamic acid accord | Tranexamic acid stragen | Tranexamsyra ebb | Tranexamsyra Orifarm | Tranon;
 - (SG) Singapore: Cyklokapron;
 - (SI) Slovenia: Cyklokapron | Exacyl | Medsamic | Traxidot | Trenolk | Ugurol;
 - (SK) Slovakia: Exacyl | Medsamic | Trenolk;
 - (SR) Suriname: Cyklokapron | Tranexaminezuur | Tranexaminezuur Mylan;
 - (TH) Thailand: Axamin;
 - (TN) Tunisia: Exacyl | Frenolyse;
 - (TR) Turkey: Herajit | Transamine;
 - (TW) Taiwan: Biosaren | Rasemin | Tranexam | Tranexamic | Tranexic | Transamin | Transamin s

- | Zecolin | Zucerin;
- (UA) Ukraine: Acemik | Atraxan | Cyklokapron | Cyklokapron zdorovj | Euronex | Gemaxam | Gemotran | Hemoactiv | Maxitran | Nexaar | Tranexam | Trenaxa | Vidanol;
- (UG) Uganda: Kapron | Plasminex | Ro tamic | Texakind | Tranlok | Trenaxa | Xenamex;
- (UY) Uruguay: Acido tranexamico blau farmaceutica | Tranexia | Transamina;
- (VE) Venezuela, Bolivarian Republic of: Biotranex | Ciclokan | Ciclokapron | Hemotran | Tarexa | Xanextra;
- (VN) Viet Nam: Bfs tranexamic | Duhemos | Tranexamic vmg;
- (ZA) South Africa: Cyklokapron | Cynex | Emkapron | Morwak | Tranecid | Tranexamic acid 100 mg/ml pharmc | Tranic | Tranmenxio iv;
- (ZW) Zimbabwe: Cyclokapron

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